Charge and the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain charge interactions and trace an electric car's energy transfers.

SUCCESS CRITERIA - I can...

1. Predict attraction or repulsion from charge signs.
2. Distinguish charge from energy.
3. Trace energy from a battery to moving wheels.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An ideal blackbody has peak wavelength **966.7 nm**.
1. Use λ_{max} $T = \mathbf{2.90 \times 10^{6}\ nm \cdot K}$.
2. **Estimate temperature**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $E_{electrical} = E_{mechanical} + E_{thermal}$ + other transfers. Energy accounting for a motor with no net energy storage. A battery transfers energy; charge already exists in the wires.

Worked reasoning: Rubbing can transfer electrons between materials. If one isolated object gains charge **-3** units and another loses the same electrons, their charge changes add to zero.

Your turn: Draw before-and-after charge accounts for both objects. Distinguish transfer from creation of charge.

Label charges and separation.

Object A Object B

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw force arrows for two positive charges, then for a positive and a negative charge.



\+ +

Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

During a small classroom car test, the battery transfers **120 J** and the car gains **30 J** of translational kinetic energy. Its height stays fixed. **Complete the energy account**, find energy in other forms and **explain why the circuit's charge** is not used up.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

Charge and the electric car

antocciphysics

Continue down each column.

05 Explain your reasoning

A student says the motor uses up the current so less returns to the battery. **Explain the mistake** for a steady series circuit.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Three instantaneous displacements are **+4 mm**, **-6 mm** and **-8 mm**. **Find their sum** and **explain why one displacement value** does not find total wave energy. **Show the signed contributions** and their sum.

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3

07 Exit check

How can an initially neutral object be attracted to a charged object without having a net charge?



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

An initially neutral pair separates with charges **+8 nC** and **-8 nC**. One then shares all its net charge equally with an identical initially neutral conductor. **Find the three final charges**.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3